

SRM VEC
DEPARTMENT OF AI&DS
Semester End Examination
(Regulations 2023)
AD3445 – AI1

Branch: B.Tech | Date: 01.03.2026 | Max. Marks: 100 | Duration: 3 Hrs

Answer ALL Questions
(100 Marks)

PART A — (10 × 2 = 20 marks)

1. Define a Rational Agent and list the four factors that depend on the measure of rationality. [CO1] [BTL1-Remember] (2)
2. Differentiate between Breadth-First Search (BFS) and Depth-First Search (DFS) in terms of completeness and optimality. [CO1] [BTL2-Understand] (2)
3. Explain the concept of 'Heuristic Function' with an example related to the 8-puzzle problem. [CO1] [BTL2-Understand] (2)
4. What is the purpose of Alpha-Beta pruning in game tree search? [CO2] [BTL1-Remember] (2)
5. Define Constraint Satisfaction Problem (CSP) and list its three components. [CO2] [BTL1-Remember] (2)
6. Convert the following sentence into First-Order Logic (FOL): 'Every student who takes AI loves programming.' [CO3] [BTL3-Apply] (2)
7. State the difference between Forward Chaining and Backward Chaining. [CO3] [BTL2-Understand] (2)
8. Define Bayes' Rule and explain its significance in uncertain reasoning. [CO4] [BTL2-Understand] (2)
9. What is a Markov Decision Process (MDP)? List its four components. [CO4] [BTL1-Remember] (2)
10. Define 'Information Gain' in the context of Decision Tree induction. [CO5] [BTL2-Understand] (2)

PART B — (5 × 16 = 80 marks)

- 11(a) (i) Explain the A* Search algorithm in detail. Prove that A* is optimal if the heuristic is admissible. (ii) Compare and contrast Informed search and Uninformed search strategies with suitable examples.

OR

- (b) (i) Describe the Hill Climbing search algorithm. Explain the problems of Local Maxima, Plateaus, and Ridges. (ii) Illustrate the working of Simulated Annealing and how it overcomes the limitations of Hill Climbing.

- 12(a) (i) Explain the Minimax algorithm for game playing with a suitable game tree. (ii) Apply Alpha-Beta pruning on a 3-level deep game tree and demonstrate how branches are pruned.

OR

- (b) (i) Formulate the Map Coloring problem as a Constraint Satisfaction Problem (CSP). (ii) Explain the Backtracking Search algorithm for CSPs and discuss the 'Minimum Remaining Values' (MRV) heuristic.

- 13(a) (i) Explain the process of Unification in First-Order Logic with an example. (ii) Describe the steps to convert a First-Order Logic formula into Conjunctive Normal Form (CNF).

OR

- (b) (i) Explain the Wumpus World environment. Discuss the role of a knowledge-based agent in navigating this environment. (ii) Demonstrate the Resolution principle in Propositional Logic with a proof by contradiction.

- 14(a) (i) Define Bayesian Networks. Explain how the joint probability distribution is represented in a Bayesian Network. (ii) Given a network: Burglary(B) and Earthquake(E) cause Alarm(A). Alarm causes MaryCalls(M) and JohnCalls(J). Explain the conditional independence relationships in this network.

OR

- (b) (i) Explain the Hidden Markov Model (HMM) and its three fundamental problems (Likelihood, Decoding, Learning). (ii) Describe the concept of Utility functions and how they are used in decision making under

uncertainty.

15(a) (i) Elaborate on the ID3 algorithm for Decision Tree construction. Explain the mathematical formula for Entropy. (ii) Discuss the concept of Overfitting in Machine Learning and suggest two methods to prevent it.

OR

(b) (i) Explain the Reinforcement Learning framework (Agent, Environment, State, Action, Reward). (ii) Describe the Q-Learning algorithm and explain the Q-table update rule.

Faculty In-Charge

Verified by

HOD